

BANKING LOAN PORTFOLIO & CREDIT RISK ANALYTICS

Project Report

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Course: Data Analytics

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1. Introduction

Banks generate large volumes of data related to customers, loan applications, loans, payments, credit history, branches, collateral, loan status, and recovery. Analyzing this information helps financial institutions understand portfolio performance, customer risk, overdue exposure, default behaviour, collateral coverage, and recovery performance. This project develops a Banking Loan Portfolio & Credit Risk Analytics solution using Python, SQL, and Power BI. The analysis combines data preparation, business calculations, statistical analysis, risk analysis, and interactive dashboard reporting to support data-driven banking decisions.

2. Problem Statement

A bank wants to evaluate the health of its loan portfolio and identify factors associated with loan default risk. The project focuses on loan approval rates, default rates, overdue accounts, loan aging, customer risk segments, collateral coverage, and recovery performance. It also examines how income, credit score, loan amount, and loan-to-value ratio are associated with default risk. The findings are intended to be presented through statistical analysis, visual summaries, and an interactive Power BI dashboard.

3. Objectives

- Import, clean, validate, and prepare the nine banking datasets using Python.
- Create a structured relational banking database using MySQL.
- Create calculated fields and perform loan portfolio and credit-risk analysis using SQL.
- Analyze loan approval, default, overdue, aging, collateral, LTV, and recovery performance.
- Create customer and loan risk categories using business-oriented rules.
- Perform descriptive, comparative, correlation, and risk analysis using Python.
- Develop an interactive Power BI dashboard with KPIs, charts, slicers, and cross-filtering.
- Identify business insights and support data-driven lending, monitoring, and recovery decisions.

4. Dataset Description

The project uses nine banking datasets representing major entities in the loan lifecycle. The task sheet specifies the following tables:

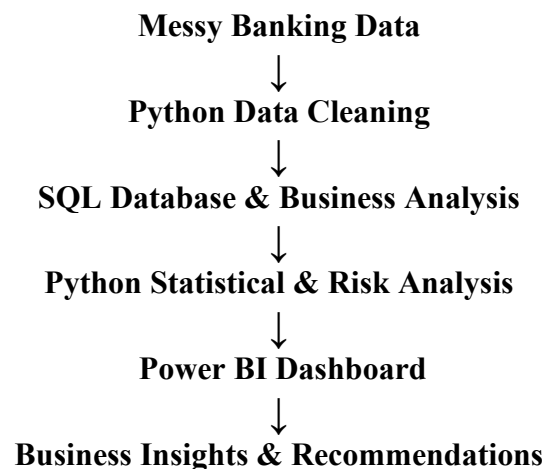
No.	Banking Table	Purpose
1	Customers	Customer information used for portfolio and risk analysis
2	Loan Applications	Application and approval-related information
3	Loans	Loan-level financial and status information
4	Loan Payments	Payment and repayment information
5	Credit History	Credit-related information for risk analysis
6	Branches	Branch-level portfolio and performance analysis
7	Collateral	Collateral value and loan-to-value analysis
8	Loan Status History	Loan status and status progression
9	Loan Recovery	Defaulted and recovered amount analysis

5. Tools and Technologies Used

Tool	Purpose
Python	Data cleaning, preparation, descriptive statistics, statistical analysis, and risk analysis
MySQL Workbench	Database creation, calculated fields, relational data management, and SQL business analysis
Microsoft Power BI	Data modeling, DAX KPI measures, interactive visualization, and dashboard reporting

6. Project Workflow

The project follows the end-to-end workflow defined in the banking task sheet:



7. Project Implementation

7.1 Data Cleaning & Preparation Using Python

Python is used in the first phase to prepare the nine banking datasets for reliable downstream analysis. The cleaning process focuses on data structure, duplicates, missing values, text consistency, dates, numeric data types, and banking-data validation.

- Import all nine datasets into separate Pandas DataFrames.
- Review row counts, column counts, data types, sample records, and basic summaries.
- Identify duplicate records and remove exact duplicates where appropriate.
- Calculate and handle missing values in important numeric and categorical fields.
- Standardize text values such as loan status and payment status.
- Convert mixed date fields into consistent datetime formats.
- Convert financial fields such as annual income, loan amount, interest rate, payment amount, recovery amount, and collateral value into numeric formats.
- Validate age, credit score, loan amount, interest rate, payment amount, recovery amount, overdue days, and loan status for invalid values.

The screenshot shows a JupyterLab window titled 'banking' with a 'Last Checkpoint: 5 days ago' status. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar. A code cell is active, showing the following code:

```
[18]: customers.info()
      loan_applications.info()
      loans.info()
      loan_payments.info()
      credit_history.info()
      branches.info()
      collateral.info()
      loan_status_history.info()
      loan_recovery.info()
```

The output of the code is displayed below the cell:

```
<class 'pandas.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   customer_id           1200 non-null  int64  
 1   first_name            1200 non-null  str     
 2   last_name             1200 non-null  str     
 3   gender                1200 non-null  str     
 4   age                   1200 non-null  int64  
 5   date_of_birth         1200 non-null  datetime64[us]
 6   marital_status        1200 non-null  str     
 7   occupation            1200 non-null  str     
 8   annual_income         1199 non-null  float64 
 9   monthly_income        1199 non-null  float64 
10   city                  1200 non-null  str
```

7.2 Database Development Using MySQL

The cleaned datasets are imported into MySQL and organized as a relational banking database. Primary keys, foreign keys, data types, and constraints are applied to support data integrity. The database is then used for calculated fields and business-oriented SQL analysis.

- Create the nine banking tables.
- Apply primary keys and foreign key relationships.
- Import and validate the cleaned datasets.
- Verify row counts and table relationships.
- Create a consolidated loan-level view by joining customer, application, loan, payment, and status information.

7.3 SQL Calculated Fields & Business Analysis

All specified business calculations and calculated categories are created in SQL rather than Python. These calculations provide a consistent foundation for portfolio and credit-risk analysis.

- Create age groups: 18–30, 31–45, 46–60, and 60+.
- Create Low, Medium, and High income groups.
- Create Low, Medium, and High credit-score categories.
- Create Low, Medium, and High loan-amount categories.
- Create overdue buckets: Current, 1–30, 31–60, 61–90, 91–180, and 180+ days.
- Calculate Loan-to-Income (LTI) = $\text{Loan Amount} \div \text{Annual Income}$.
- Calculate Loan-to-Value (LTV) = $\text{Loan Amount} \div \text{Collateral Value}$.
- Create Low Risk, Medium Risk, and High Risk categories.
- Analyze approval rate and default rate by relevant customer and loan segments.
- Summarize customer-level applications, approved loans, loan amount, outstanding balance, overdue loans, and defaults.
- Analyze overdue exposure, collateral coverage, recovery performance, monthly/yearly trends, and rankings.

```

134
135 • select loan_type, count(loan_id) as total_loans,
136       count(case
137         when loan_status = 'Defaulted' then 1 end) as default_loans,
138       round((
139         count(case
140           when loan_status = 'Defaulted' then 1 end) * 1.0 / count(loan_id)) * 100, 2) as default_rate
141     from banking.loans
142    group by loan_type;
143
144 -- Task 21 - Customer Loan Summary. How much loan exposure does each customer have?--
145 • select c.customer_id, c.first_name, count(distinct a.application_id) as total_application, sum(l.loan_amount) as total_loan_amount, sum(l.outstanding_balance)
146       count(case
147         when a.approval_status = 'Approved' then 1 end) as Approved_Applications,
148       count(case
149         when loan_status = 'Defaulted' then 1 end) as defaulted_loans,
150       count(distinct case
151         when p.days_overdue > 0 then l.loan_id end) as Overdue_Loans

```

8. Python Statistical & Risk Analysis

After SQL calculations and business analysis, Python is used for statistical and risk-focused analysis. This phase is intended to identify differences between defaulted and non-defaulted loans and relationships among key financial variables.

- Calculate mean, median, minimum, maximum, and standard deviation for income, credit score, loan amount, EMI, overdue days, and recovery amount.
- Compare defaulted and non-defaulted loans using income, credit score, loan amount, EMI, and outstanding balance.
- Compare default rates across income groups.
- Compare default rates across credit-score categories.
- Compare default rates across loan-amount categories.
- Compare Low Risk, Medium Risk, and High Risk segments using customer count, loan exposure, default count, and default rate.
- Perform correlation analysis for income, credit score, loan amount, interest rate, EMI, outstanding balance, overdue days, and collateral value.

```

[55]: # Merge customer information with loans
risk_data = loans.merge(
    customers[['customer_id', 'annual_income', 'credit_score']],
    on='customer_id',
    how='left'
)

# Calculate LTI
risk_data['LTI'] = (
    risk_data['loan_amount'] / risk_data['annual_income']
).round(2)

# Create Risk Segment
risk_data['risk_segment'] = 'High Risk'

risk_data.loc[
    (risk_data['credit_score'] >= 700) &
    (risk_data['LTI'] <= 3),
    'risk_segment'
] = 'Low Risk'

risk_data.loc[
    (risk_data['credit_score'] >= 600) &
    (risk_data['LTI'] <= 5) &
    ~((
        (risk_data['credit_score'] >= 700) &
        (risk_data['LTI'] <= 3)
    )),
    'risk_segment'

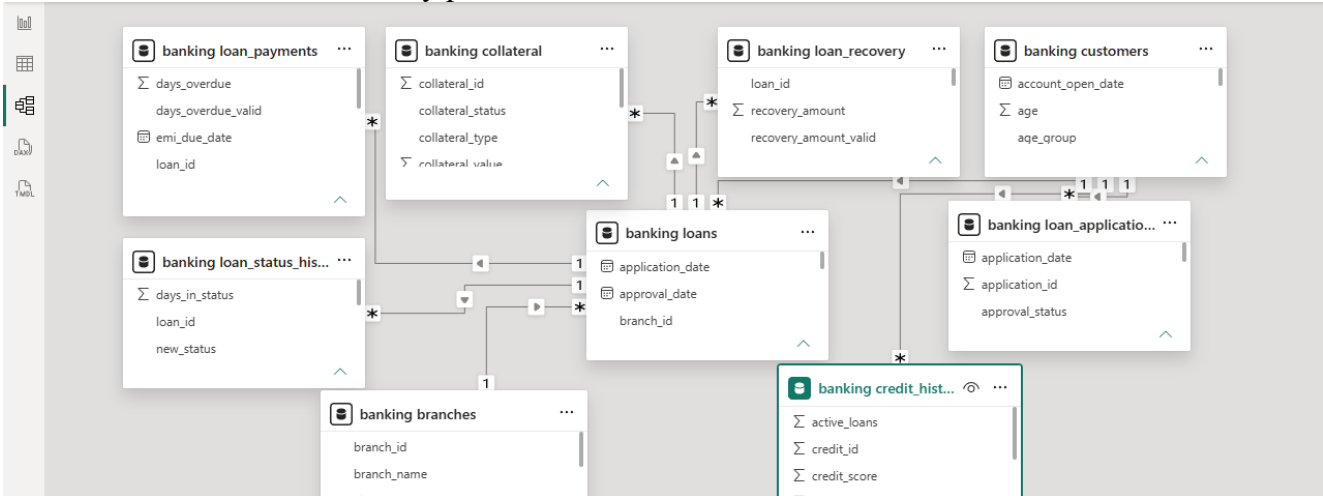
```

9. Power BI Implementation

Power BI is used to connect to the SQL data model and present the results through an interactive banking dashboard. The dashboard is designed to provide management-level visibility into portfolio size, approvals, defaults, overdue exposure, aging, risk segments, collateral coverage, and recovery.

9.1 Data Modeling

- Connect Power BI to SQL and load the required banking tables.
- Create and validate relationships between related tables.
- Organize the model for accurate reporting.
- Create DAX measures for key portfolio and risk KPIs.

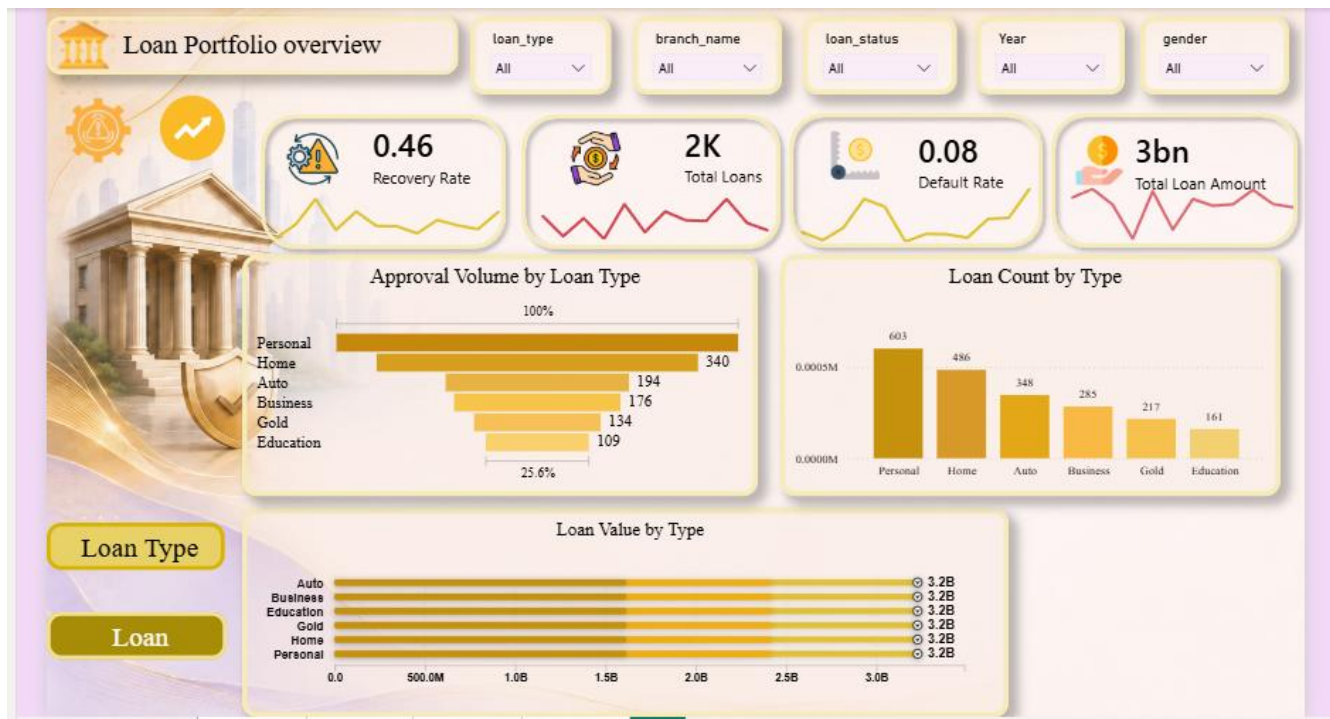


9.2 Key DAX KPIs

- Total Loans
- Loan Amount
- Outstanding Balance
- Approval Rate
- Default Rate
- Overdue Amount
- Recovery Rate

9.3 Dashboard Pages / Analysis Areas

Dashboard Area	Purpose
Portfolio Overview	Overall loan portfolio, loan status, loan type, loan amount, and portfolio KPIs
Approval & Default Risk	Approval/default rates by loan type, income group, credit score, customer segment, and risk category
Aging & Recovery	Overdue buckets, outstanding amount, defaulted amount, recovered amount, recovery rate, and LTV categories
Portfolio Trends	Monthly applications, approvals, defaults, approval rate, and default rate
Interactive Analysis	Slicers, cross-filtering, and tooltips for Loan Type, Loan Status, Risk Category, Income Group, Credit Score, Loan Amount, Branch, and Year



10. Business Insights

The uploaded banking task sheet defines the analyses and KPIs to be calculated, but it does not provide the final dataset results. Therefore, numerical banking findings are not stated here. The completed analysis should populate this section with verified results from the SQL queries, Python analysis, and Power BI dashboard.

- Portfolio insight: summarize total loans, total loan amount, and outstanding exposure.
- Approval insight: identify approval rate and the strongest/weakest approval segments.
- Default insight: identify the loan types and customer segments with the highest default rates.

- Risk insight: compare Low, Medium, and High Risk segments using exposure and default rate.
- Aging insight: identify the largest overdue buckets and their portfolio exposure.
- Collateral insight: identify higher-LTV categories and their associated default rates.
- Recovery insight: compare recovered amount and recovery rate across loan types and branches.
- Trend insight: identify meaningful monthly or yearly changes in applications, approvals, overdue loans, and defaults.

11. Business Recommendations

Recommendations should be finalized from the verified analytical results. Based on the analysis framework specified in the task sheet, the following decision areas should be addressed:

- Strengthen monitoring and underwriting for segments showing consistently higher default rates.
- Use credit-score, income, loan-amount, LTI, and LTV indicators as supporting factors in risk assessment.
- Prioritize overdue accounts according to aging buckets and outstanding exposure.
- Review higher-LTV loans for weaker collateral coverage and increased monitoring requirements.
- Improve recovery strategies by comparing recovery performance across branches and loan types.
- Use customer and risk segmentation to support differentiated portfolio monitoring.
- Track monthly and yearly approval and default trends to support proactive portfolio management.
- Monitor Power BI KPIs regularly so management can identify emerging portfolio risks promptly.

12. Challenges Faced

- Cleaning and standardizing multiple banking datasets.
- Handling duplicates and missing values without compromising analytical quality.
- Maintaining consistency across financial, categorical, and date fields.
- Understanding and implementing relationships among nine relational banking tables.
- Writing SQL queries involving multiple tables and calculated fields.
- Creating meaningful risk and overdue categories.
- Performing customer- and loan-level statistical analysis.
- Connecting the SQL model to Power BI and validating relationships and measures.
- Designing a dashboard that communicates credit-risk information clearly and interactively.

13. Conclusion

The Banking Loan Portfolio & Credit Risk Analytics project demonstrates an end-to-end data analytics workflow for banking decision support. Python is used for data cleaning, preparation, and statistical/risk analysis; MySQL is used for relational database development, calculated fields, and business analysis; and Power BI is used for data modeling, KPI development, and interactive dashboard reporting. The solution is designed to provide a consolidated view of loan portfolio performance, approval and default behaviour, overdue aging, customer risk, collateral coverage, and recovery performance. Once the actual analytical outputs are populated, the dashboard can support timely and data-driven lending, monitoring, and recovery decisions.